Le Hong Phuc

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EDUCATION

University of Rochester — **Rochester, New York**

- B.S. in Computer Science and Mathematics. GPA: **3.98 / 4.00** — Expected May 2029
- Coursework: AI, Data Structures & Algorithms, Computation and Formal Systems, Probability, Linear Algebra

EXPERIENCE

Research Assistant — **September 2026 – Present**

HeLab — University of Rochester

- Developed scGPT brain cell-type annotation pipelines using PyTorch, Scanpy, and AnnData, comparing fine-tuning with pretrained embedding-based reference mapping.
- Fine-tuned scGPT and evaluated **848,701 brain nuclei** across held-out datasets, achieving **92.76% accuracy and 88.08% macro-F1** on unseen donors; outperformed reference-mapping baselines in both metrics.
- Engineered reference mapping with memory-efficient data streaming, resumable embeddings, and GPU-accelerated nearest-neighbor search; achieved faster evaluation and better performance on challenging cell types despite lower overall accuracy and macro-F1.

Software Engineer Intern — **May 2026 – July 2026**

VecML — Remote

- Developed a Vietnamese RAG evaluation benchmark using **300+ domain documents** and curated question-answer pairs to assess retrieval and generation performance.
- Curated financial test datasets in StarRocks and configured Supabase PostgreSQL; accessed Linux VMs via SSH and executed code within Docker containers.

PROJECTS

Stella - Project Planning Platform | *FastAPI, PostgreSQL, SQLAlchemy, Pydantic AI, pgvector*

- Collaborated with a team to build a full-stack AI project planning platform that transforms vague ideas into structured roadmaps through adaptive Q&As, goal generation, milestone planning, and chat assistance.
- Developed the FastAPI backend, optimized database queries through SQLAlchemy eager loading to eliminate N+1 queries, and built a ReAct-style AI agent with project retrieval and web search tools.
- Co-designed and implemented semantic project memory using pgvector embeddings and project-scoped cosine similarity search, alongside rolling summarization to preserve context across long-running conversations.

AWS Infrastructure Copilot | *TypeScript, MCP, AWS Lambda, Next.js, Zod*

- Built a TypeScript MCP server exposing **7 read-only tools across 6 AWS services**, enabling Claude to answer natural-language infrastructure queries using AWS SDK v3 and Zod-validated tool schemas.
- Deployed an MCP backend to AWS Lambda via Streamable HTTP and built a streaming Next.js chat interface secured with least-privilege IAM permissions.

Active Learning for Bottle Defect Detection | *Python, PyTorch, ResNet-18, OpenCV, scikit-learn*

- Developed a ResNet-18 defect classifier using OpenCV-generated synthetic defects and uncertainty-based active learning to reduce manual labeling.
- Achieved **93.2% accuracy and 85% F1 score**, reaching **~90%** accuracy with **50% of real training labels** and consistently outperforming random sampling.

TECHNICAL TRAINING

- **CodePath AI 201 - Applications of AI Engineering** (Github)**:** Developed a RAG pipeline with multi-format document ingestion, hybrid retrieval (dense vectors + BM25), reciprocal rank fusion, and cross-encoder reranking.
- **Backend Development Projects** (Github): Built REST APIs with Node.js, Express, and PostgreSQL; practiced containerization and AWS EC2 deployment using Docker, Terraform, Ansible, and GitHub Actions.

HONORS & LEADERSHIP

- Silver Prize, ICPC North America Northeast Regional Contest — 2025
- First Prize, Hanoi Excellent Math Student Selection Exam — 2022
- USAII Hackathon Finalist — 2026
- Workshop Leader (Data Structures & Algorithms) and Study Group Leader (Discrete Mathematics)

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, C, TypeScript, JavaScript
- **Frameworks & AI/ML:** PyTorch, scikit-learn, Scanpy, AnnData, FAISS, FastAPI, React, Next.js, Pydantic AI
- **Tools & Databases:** AWS, Docker, Linux, Git, PostgreSQL, pgvector, Slurm, GitHub Actions, Terraform